

Ross Kelso

Glasgow, Scotland

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Experienced software developer with 5+ years experience in full-stack development, data engineering, and AI.

Skilled in rapid build out of efficient data analysis/visualisation systems in `c#`, `python`, and `javascript`.

Driven to explore emerging technologies and frameworks.

First class degree in Computer science with a background in industrial software, machine learning, and computer vision.

Technical Skills

Languages & Frameworks: `c#`, `python`, `javascript`, `.net`, `react`, `sql`

Technologies: `elasticsearch`, `git`, `azure`, `docker`, `kubernetes`, `tensorflow`, `ollama`, `nodejs`, `windows`, `linux`, `ci`

Concepts: Machine Learning, Data Engineering, Cloud Architecture, Computer Vision

Work Experience

Intelligent Plant - Software Developer (2021-present), Intern (2017-2021)

Intelligent Plant specialises in providing data storage, analysis and visualisation solutions for the energy industry.

Lead developer for Sequence of Events Explorer `c#` `javascript` `html` `css` `elasticsearch` `ci`

- Developed the application from inception to maturity, including design, development, deployment, and maintenance resulting in it being adopted by multiple major industrial clients.
- Enhanced user experience by developing an intuitive, and powerful frontend for advanced data queries.
- Improved code quality and maintainability by managing developers through rigorous code reviews and enforcing best practices.

Developer for Alarm Analysis and Intelligent Plant Historian `c#` `elasticsearch`

- Developed and maintained a backend for performing complex data aggregations.
- Worked with clients directly to facilitate on-site installations.
- Upgraded the historian across several major versions of Elastic Search. This more than doubled the size of a query that can be performed in 1s and increased the size of a query that can be performed in 5s by about 10 times.

Developer for Gestalt (PnID & Trend) `c#` `sql` `javascript`

- Lead the project to decouple the Gestalt web app from our data API, resulting in smoother deployments for Gestalt and our API.

- Designed and implemented a database model to replace an existing file system dependency paving the way for migration to App Services and zero downtime.

Lead developer for the Industrial App Store API Python Client `python` `pandas` `django`

- Designed and implemented a Python SDK to simplify access to the Industrial App Store API.
- Developed comprehensive documentation and example use cases for ease of adoption by other developers.
- Ensured API client was robust and extensible to support future enhancements and additional features.

Experimental projects involving the application of AI and large language models to our software `python` `docker` `tensorflow` `ollama`

- Deployed internal AI inference APIs.
- Researched and developed proof-of-concept applications integrating LLMs for industrial automation.
- Explored use cases such as natural language querying of industrial data.

Lead developer for Power BI Industrial Connector `c#` `power query m` `react`

- Developed a high-performance data connector to enable seamless industrial data access in Power BI.
- Optimized data retrieval and transformation processes to improve query efficiency and responsiveness.

Optimized the Excel Data Query Add-in, reducing query time from minutes to seconds. `c#`

Education

University of St Andrews - Computer Science MSci First Class (2016-2021)

Deans' List, for academic excellence 2017/18, 2018/19, 2019/20

Masters Thesis

Synthetic GPS Traces for Privacy Protection `python` `tensorflow` `pandas` `numpy`

- Using Markov Models and generative AI to synthesize realistic GPS traces.
- Evaluated model accuracy and effectiveness in preserving statistical properties of real-world movement data.

Senior Honours Project

Computer Vision in Cue Sports `opencv` `python`

- Designed an algorithmic computer vision system to analyze pool table positions without the need for markers.
- Implemented perspective transformation and object detection techniques for accurate ball position detection.

Junior Honours Project

Group software engineering project to create an encrypted wireless network of embedded devices `c`

`nodejs` `python` `git`

- Applied low-level networking principles to develop a secure, distributed communication protocol.
- Implemented encryption mechanisms for secure data transmission.
- Worked in an Agile environment, utilizing SCRUM methodologies for iterative development and collaboration.

Other Modules

- Algorithms and data structures `java`
- Artificial intelligence and machine learning `python` `java`
- Information Visualisation `python`
- Operating systems `c`
- Advanced programming projects `haskell` `python`

Personal Interests

Starfinder/D&D Game Master (2024-present)

- Leadership and Organization: Coordinated and led a group of players through a immersive and engaging story, ensuring a cohesive and enjoyable experience for all participants.
- Creative Problem-Solving: Designed and implemented challenging scenarios, puzzles, and encounters that required players to think critically and strategically to overcome.
- Adaptability and Improvisation: Improvised and responded to unexpected events, player choices, or rule interpretations, using creative problem-solving and diplomacy to resolve conflicts and maintain a positive game environment.

Pool Society Vice-president, Social Rep, Publicity and Fundraising Rep (2017-2021)

- Managed day-to-day operations, including equipment maintenance, weekly sessions and recruiting new members.
- Promoted the society through social media channels and organized fundraisers, including live streams of semesterly Finals Day events.
- Represented the university at several pool competitions, including 5 BUCS tournaments.